Mex Grid Construction

Time limit: 1.00 s
Memory limit: 512 MB

CSES 3419 | Introductory Problems | https://cses.fi/problemset/task/3419

Your task is to construct an $n \times n$ grid where each square has the smallest nonnegative integer that does not appear to the left on the same row or above on the same column.

Input

The only line has an integer n.

Output

Print the grid according to the example.

Constraints

- $1 \le n \le 100$

Example

Input
5

Output
0 1 2 3 4 1 0 3 2 5 2 3 0 1 6 3 2 1 0 7 4 5 6 7 0